

Python Basics - Lecture 1 | Practical Quiz

10 Coding Problems with Solutions

PROBLEMS TO SOLVE

1. Write a program that takes a number from user input and prints if it's even or odd.

2. What will this code print? Debug and explain the output.

```
x = "5"  
y = "3"  
print(x + y)
```

3. Write code to take a student's name and marks, then display their name and marks + 5.

4. What is the output? $(y + z) * x = ?$

```
x = 2  
y = 10 / 5  
z = 3 * 2
```

5. Write a program using floor division ($//$) to find how many complete groups of 3 are in 17.

6. Fix the error in this code:

```
age = input("Age: ")  
next_year_age = age + 1  
print(next_year_age)
```

7. Write code to calculate the area of a rectangle using user input (length $\times$ width).

8. What will be the data type of result? Predict the output.

```
a = 10  
b = 3  
result = a ** b / 2  
print(type(result), result)
```

9. Write a program that takes two numbers and swaps their values, then prints both.

10. Write code that takes a price and applies a 20% discount, then prints the final price.

SOLUTIONS

Problem 1:

1. Write a program that takes a number from user input and prints if it's even or odd.

Code:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Explanation: Use modulus (%) to get remainder. If remainder is 0, it's even; otherwise odd.

Problem 2:

2. What will this code print? Debug and explain the output.

```
x = "5"
y = "3"
print(x + y)
```

Output: 53

Explanation: Since x and y are strings (not integers), the + operator concatenates them instead of adding. Result is "53" not 8. To get 8, convert to int first: print(int(x) + int(y))

Problem 3:

3. Write code to take a student's name and marks, then display their name and marks + 5.

Code:

```
name = input("Enter name: ")
marks = int(input("Enter marks: "))
print("Student:", name)
print("Updated marks:", marks + 5)
```

Example output: Student: Anant
Updated marks: 23

Problem 4:

4. What is the output? $(y + z) * x = ?$
 $x = 2$

y = 10 / 5
z = 3 * 2

Step by step:

y = 10 / 5 = 2.0 (division returns float)

z = 3 * 2 = 6

(y + z) * x = (2.0 + 6) * 2 = 8.0 * 2 = 16.0

Output: 16.0

Problem 5:

5. Write a program using floor division (//) to find how many complete groups of 3 are in 17.

Code:

```
num = 17
```

```
groups = num // 3
```

```
print(f"Complete groups of 3: {groups}")
```

Output: Complete groups of 3: 5

Note: 17 // 3 = 5 (not 5.67)

Problem 6:

6. Fix the error in this code:

```
age = input("Age: ")
```

```
next_year_age = age + 1
```

```
print(next_year_age)
```

Fixed Code:

```
age = int(input("Age: "))
```

```
next_year_age = age + 1
```

```
print(next_year_age)
```

Error: input() returns string. Can't add 1 to a string. Solution: Convert using int()

Problem 7:

7. Write code to calculate the area of a rectangle using user input (length × width).

Code:

```
length = float(input("Enter length: "))
```

```
width = float(input("Enter width: "))
```

```
area = length * width
```

```
print(f"Area: {area}")
```

Note: Use float() to allow decimal inputs. Output example: Area: 12.5

Problem 8:

8. What will be the data type of result? Predict the output.

a = 10

b = 3

result = a ** b / 2

print(type(result), result)

Calculation:

$a ** b = 10 ** 3 = 1000$

$1000 / 2 = 500.0$ (division returns float)

Output: <class 'float'> 500.0

Key point: Division (/) always returns float

Problem 9:

9. Write a program that takes two numbers and swaps their values, then prints both.

Code:

a = int(input("First number: "))

b = int(input("Second number: "))

a, b = b, a

print(f"After swap: a={a}, b={b}")

Example: Input: 5, 10 → Output: a=10, b=5

Note: Python allows tuple unpacking for easy swapping

Problem 10:

10. Write code that takes a price and applies a 20% discount, then prints the final price.

Code:

price = float(input("Enter price: "))

discount = 20

final_price = price - (price * discount / 100)

print(f"Final price: {final_price}")

Or simpler: **final_price = price * 0.8**

Example: Input: 100 → Output: Final price: 80.0